Juan Belieni de Castro Araujo

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Education

Getulio Vargas Foundation, School of Applied Mathematics — Rio de Janeiro, Brazil
MSc in Applied Mathematics and Data Science — *2025 — Present*
- Researching topics in Machine Learning and Artificial Intelligence, with emphasis on LLMs, AI Safety and GNNs.

Getulio Vargas Foundation, School of Applied Mathematics — Rio de Janeiro, Brazil
BSc in Data Science and Artificial Intelligence — *2021 — 2024*
- Curriculum spanning Applied Mathematics, Statistics, Deep Learning, and Software Engineering.
- Teaching Assistant for Machine Learning, Natural Language Processing, and Reinforcement Learning.
- Presented two works at CNMAC 2024: "SRGNN: Simple Recurrent GNN," proposing optimizations to recurrent GNN architectures, and ChoveuRIO, a rainfall visualization platform for Rio de Janeiro.

Federal Institute of Southern Minas Gerais — Minas Gerais, Brazil
Technical High School in Information Technology — *2018 — 2020*
- Completed technical program focused on algorithms, hardware, and software development.
- Participated in software engineering and web development projects.

Experience

Teaching Assistant — Apr 2025
ML4Good — *Colombia*
- Delivered technical workshops on machine learning, interpretability, and reinforcement learning.
- Mentored participants on project development, providing guidance on implementation and research methodology.

Researcher — Jan 2024 — Present
Getulio Vargas Foundation, Green AI Lab — *Rio de Janeiro, Brazil*
- Conducting research on GNNs and LLMs, resulting in three publications so far.
- Implemented deep learning architectures in PyTorch, working with large-scale models and HPC infrastructure.

Data Science Intern — Feb 2023 — May 2023
Quantum Finance — *Rio de Janeiro, Brazil*
- Built automated ETL pipelines using Python and SQL to scrape and structure financial data.
- Applied NLP techniques to analyze unstructured text data in support of quantitative financial modeling.
- Optimized data processing workflows with Pandas and NumPy, improving data availability for the analytics team.

Software Engineer — Jan 2021 — Feb 2023
Social Comics Entretenimento — *Remote*
- Built and maintained full-stack applications using Vue.js, TypeScript, HTML, and CSS.
- Developed cross-platform mobile applications with Flutter, ensuring consistent performance on iOS and Android.
- Collaborated in an agile environment using Git for version control and participating in code reviews.

Publications

Orthogonal Gradient Projection for Continual LLM Unlearning — RSI Workshop @ ICLR 2026
- Proposed Orthogonal Negative Preference Optimization, which handles sequential machine unlearning requests in LLMs by projecting gradient updates onto the orthogonal complement of prior unlearning subspaces, achieving a better forgetting-utility trade-off on TOFU than existing methods.

Math-PT: A Math Reasoning Benchmark for European and Brazilian Portuguese — PROPOR 2026
- Introduced a dataset of 1,729 Portuguese-language math problems drawn from Olympiads and exams in Brazil and Portugal, providing a benchmark for evaluating LLMs on non-English mathematical reasoning.

Cooperative Sheaf Neural Networks — ICLR 2026
- Extended sheaf neural networks to directed graphs with in- and out-degree Laplacians, enabling nodes to independently control information propagation; proved selective long-range attention properties that alleviate oversquashing.

Selected Projects

More at belieni.me/projects.

Watermark Stealing Attacks Replication Aug 2025

- Replicated experiments from the "Watermark Stealing" paper as a capstone project.
- The goal was to demonstrate vulnerabilities in statistical text watermarking schemes for LLMs.
- Implemented attack vectors to extract and circumvent watermarks using Python.

Multilingual Sparse Autoencoders Analysis Jan 2025

- Conducted an exploratory analysis and visualization of SAE feature robustness across languages to assess failure modes in multilingual models.

ACDC (Automated Circuit Discovery) Replication Jul 2024

- Replicated the "Towards Automated Circuit Discovery" paper, implementing algorithms to identify functional circuits within Transformer models.
- Applied interpretability techniques to reverse-engineer model behaviors, connecting theoretical alignment work to concrete implementation.

Extracurricular Activities

AI Security Bootcamp Aug 2025

- Intensive 4-week program on security fundamentals for AI systems, including adversarial robustness and alignment.

CNMAC (Applied Mathematics Conference) Sep 2024

- Presented poster on "SRGNN: Simple Recurrent GNN," demonstrating optimizations in graph neural networks.
- Presented "ChoveuRIO", a rainfall visualization platform for Rio de Janeiro using graph-theoretic interpolation methods.

ML4Good Brazil Jul 2024

- 10-day technical bootcamp on AI alignment, interpretability and technical AI safety research.

Condor Camp Feb 2024

- 1-week intensive bootcamp on AI safety, alignment, interpretability, and governance.

Skills

- **Languages**: Python, R, C/C++, Java, JavaScript/TypeScript, HTML/CSS, Rust, SQL.
- **Tools & Infrastructure**: Docker, AWS, Git/GitHub, Linux.
- **Data & ML**: PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn, Matplotlib, Tidyverse.
- **Domains**: Machine Learning, Graph Neural Networks, Mechanistic Interpretability, AI Safety, NLP.

Awards

- **OBMEP (Brazilian Mathematics Olympiad)**: gold medalist in 2015 and 2017; medalist in 2016, 2018, and 2019.
- **OMIF (Federal Institutes Mathematics Olympiad)**: gold medalist in 2020.